EEC 325 LECTURE NOTE (1)

INTRODUCTION:

Laplace transformation is a very powerful method of solving linear differential equations. As the transient response of an electrical circuit can best be described by a linear differential equation hence, Laplace transformation finds its application in solving the transient behaviour of the electric circuits.

LAPLACE TRANSFORMATION

The time function being usually designated by lower case letter and the Laplace transformation by capital letter, the Laplace transformation of a function $f(t)$ is defined as

$$F(s) = \mathcal{L}(t) = \int_0^{\infty} f(t) e^{-st}\, dt.$$

Example 1: Laplace of a common function.

(1) Find the Laplace transform of $f(t) = a$ constant

$$\mathcal{L}[a] = \int_0^{\infty} e^{-st} \cdot a\, dt.$$

$[t=\infty,\ t=0]$.

$$\Rightarrow a\left[\frac{e^{-st}}{-s}\right]_0^{\infty} = \frac{-a}{s}\left[e^{-s\cdot\infty} - e^{s\cdot 0}\right]$$

$$\mathcal{L}[a] = \frac{-a}{s}[0-1] = \frac{a}{s}$$

$$\mathcal{L}[a] = \frac{a}{s}$$

② Find the laplace tranform of $f(t) = e^{at}$ (a constant).

$$\mathcal{L}[e^{at}] = \int_0^\infty e^{at} \cdot e^{-st}\, dt.$$

$$\Rightarrow \int_0^\infty e^{-(s-a)t}\, dt.$$

$$\Rightarrow \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty \Rightarrow -\frac{1}{s-a}\left[e^{-(s-a)\cdot\infty} - e^{-(s-a)\cdot 0}\right]$$

$$\Rightarrow \frac{1}{s-a}\left[0-1\right] = \frac{1}{s-a}$$

$$\mathcal{L}[e^{at}] = \frac{1}{s-a}.$$

③ Find the laplace transform of $f(t) = 4$.

$$\mathcal{L}[4] = \int_0^\infty 4 \cdot e^{-st}\, dt.$$

$$\Rightarrow 4\int_0^\infty e^{-st}\, dt \Rightarrow 4\left[\frac{e^{-st}}{-s}\right]_0^\infty$$

$$\Rightarrow \frac{-4}{s}\left[e^{-s\cdot\infty} - e^{-s\cdot 0}\right] \Rightarrow \frac{4}{s}\left[0-1\right] = \frac{4}{s}$$

$$\mathcal{L}[4] = \frac{4}{s}$$

(4) Find the laplace transform of $f(t) = e^{4t}$.

$$\mathcal{L}[e^{4t}] = \int_0^\infty e^{4t} \cdot e^{-st}\, dt$$

$$\Rightarrow \int_0^\infty \frac{e^{-(s-4)t}}{-(s-4)}\, dt. \Rightarrow \frac{1}{-(s-4)}\left[e^{-(s-4)\cdot\infty} - e^{-(s-4)\cdot 0}\right]$$

$$\Rightarrow \frac{1}{-(s-4)}\left[0-1\right] \Rightarrow \frac{1}{s-4}$$

$$\mathcal{L}[e^{4t}] = \frac{1}{s-4}.$$

(5) Find the laplace transform of $f(t) = -5$

$$\mathcal{L}(-5) = \int_0^\infty -5 \cdot e^{-st}\, dt.$$

$$\Rightarrow -5\left[\frac{e^{-st}}{-s}\right]_0^\infty \Rightarrow \frac{-5}{-s}\left[e^{-st}\right]_0^\infty \Rightarrow \frac{-5}{-s}\left[e^{-s\cdot\infty} - e^{-s\cdot 0}\right]$$

$$\Rightarrow \frac{-5}{-s}\left[0-1\right] = \frac{-5}{s}$$

$$\mathcal{L}[-5] = \frac{-5}{s}$$

(6) Find the Laplace Transform of $f(t) = e^{-2t}$.

$$\mathcal{L}(e^{-2t}) = \int_0^\infty e^{-2t} \cdot e^{-st}\, dt$$

$$\Rightarrow \int_0^\infty \frac{e^{-(s-2)t}}{-(s-2)}\, dt \Rightarrow \frac{1}{-(s-2)}\left[e^{-(s-2)\cdot\infty} - e^{-(s-2)\cdot 0}\right]$$

$$\Rightarrow \frac{1}{-(s-2)}\left[0-1\right] \Rightarrow \underline{\underline{\frac{1}{s-2}}}$$

$$\mathcal{L}(e^{-2t}) = \underline{\underline{\frac{1}{s-2}}}.$$

(7) Find the laplace transform of $f(t) = \sin at$.

$$\mathcal{L}[\sin at] = \int_0^\infty \sin at \cdot e^{-st}\, dt.$$

We solve by Using Integration by part

$$\sin at = \frac{e^{jat} - e^{-jat}}{2j} \quad \text{and} \quad \cos at = \frac{e^{jat} + e^{-jat}}{2}$$

$$\mathcal{L}[\sin at] = \int_0^\infty \left(\frac{e^{jat} - e^{-jat}}{2j}\right) \cdot e^{-st}$$

$$= \int_0^\infty \frac{e^{jat} \cdot e^{-st} - e^{-jat} \cdot e^{-st}}{2j}$$

$$\Rightarrow \frac{1}{2j} \int_0^\infty \left[e^{-(s-ja)t} - e^{-(s+ja)}\right] dt.$$

$$\Rightarrow \frac{1}{2j}\left[\frac{e^{-(s-ja)t}}{-(s-ja)} + \frac{e^{-(s+ja)t}}{(s+ja)}\right]_0^\infty$$

$$\text{for } t = \infty \Rightarrow \frac{1}{2j}\left[\frac{e^{-(s-ja)}}{-(s-ja)} \cdot \infty,\ \frac{e^{-(s+ja)t}}{(s+ja)} \cdot \infty\right]$$

for $t \Rightarrow 0$ $\left[\dfrac{e^{-(s-ja)}}{-(s-ja)} \cdot 0, \; \dfrac{e^{-(s+ja)t}}{(s+ja)} \cdot 0\right]$

$$\Rightarrow \frac{1}{2j}\left[0+0+\frac{1}{(s+a)}+\frac{1}{-(s-ja)}\right]$$

$$\Rightarrow \frac{1}{2j}\left[\frac{1}{(s+ja)}-\frac{1}{(s-ja)}\right] \text{ or } \left[\frac{1}{(s-ja)}-\frac{1}{(s+ja)}\right]$$

$$\Rightarrow \frac{1}{2j}\left[\frac{(s+ja)-(s-ja)}{(s-ja)(s+ja)}\right]$$

$$\Rightarrow \frac{1}{2j}\left[\frac{s+ja-s-ja}{s^2-sja+sja-ja^2}\right] = \left[\frac{2ja}{s^2+ja^2}\right]$$

$J \Rightarrow \underline{1}$

$$= \frac{1}{2j}\left[\frac{2ja}{s^2+ja^2}\right] = \frac{a}{s^2+a^2}$$

$$L(\sin at) = \frac{a}{s^2+a^2}$$

(8) Find the laplace transform of $f(t) = \cos at$.

$$L(\cos at) = \int_{t=0}^{\infty} \cos at \cdot e^{-st}\, dt.$$

From the above we know that $\cos at = \dfrac{e^{jat}+e^{-jat}}{2}$.

$$\text{Then } L[\cos at] = \int_0^{\infty} \frac{e^{jat}+e^{-jat}}{2} \cdot e^{-st}\, dt.$$

$$= \frac{1}{2}\int_0^{\infty} e^{-(s-ja)t} + e^{-(s+ja)t}\, dt.$$

$$\Rightarrow \frac{1}{2}\left[\frac{e^{-(s-ja)t}}{-(s-ja)} - \frac{e^{-(s+ja)}}{-(s+ja)}\right]_0^\infty$$

$$\Rightarrow \frac{1}{2}\left[\frac{e^{-(s-ja)\infty}}{-(s-ja)} - \frac{e^{-(s+ja)\infty}}{-(s+ja)}\right] - \left[\frac{e^{-(s-ja)\cdot 0}}{-(s-ja)} - \frac{e^{-(s+ja)\cdot 0}}{-(s+ja)}\right]$$

$$\Rightarrow \frac{1}{2}\left[0+0+\frac{1}{(s-ja)}+\frac{1}{(s+ja)}\right]$$

$$\Rightarrow \frac{1}{2}\left[\frac{1}{(s-ja)}+\frac{1}{(s+ja)}\right]$$

$$\Rightarrow \frac{1}{2}\left(\frac{(s+ja+s-ja)}{(s-ja)(s+ja)}\right) = \frac{2s}{s^2+a^2}$$

$$\Rightarrow \frac{1}{2}\left[\frac{2s}{s^2+a^2}\right] = \frac{s}{s^2+a^2}$$

$J = 1.$

$$\mathcal{L}(\cos at) = \frac{s}{s^2+a^2}.$$

(9) Find the laplace transform of $f(t) = e^{3t}$.

$$\mathcal{L}\left(e^{3t}\right) = \int_0^\infty e^{-st} \cdot e^{3t}\, dt.$$

$$= \int_{t=0}^\infty e^{-(s-3)t}\, dt.$$

$$\Rightarrow \left[\frac{e^{-(s-3)t}}{-(s-3)}\right]_{t=0}^{\infty} = -\frac{1}{(s-3)}\left[e^{-(s-3)\cdot\infty} - e^{-(s-3)0}\right]$$

$$\Rightarrow \frac{1}{-(s-3)}[0-1] \Rightarrow \frac{1}{(s-3)} = \mathcal{L}(e^{3t}) = \frac{1}{(s-3)}.$$

(10) Find the laplace transform of $f(t) = \sin wt$,

$$\mathcal{L}(\sin wt) = \int_0^\infty e^{-st} \cdot \sin wt \, dt.$$

but $\sin wt = \frac{1}{2j}\left(e^{jwt} - e^{-jwt}\right)$

$$\Rightarrow \frac{1}{2j}\left[\frac{e^{-(s-jw)t}}{-(s-jw)} - \frac{e^{-(s+jw)t}}{-(s+jw)}\right]_{t=0}^{\infty}$$

$$\Rightarrow \frac{1}{2j}\left[\frac{e^{-(s-jw)\infty}}{-(s-jw)} - \frac{e^{-(s+jw)\cdot\infty}}{-(s+jw)}\right] - \left[\frac{e^{-(s-jw)0}}{-(s-jw)} - \frac{e^{-(s+jw)0}}{-(s+jw)}\right]$$

$$\Rightarrow \frac{1}{2j}\left[\frac{1}{(s-jw)} - \frac{1}{(s+jw)}\right]$$

$$\Rightarrow \frac{1}{2j}\left[\frac{s+jw-(s-jw)}{(s-jw)(s+jw)}\right]$$

$$\Rightarrow \frac{1}{2j}\left(\frac{s-s+jw+jw}{s^2+jws-jws+jw^2}\right)$$ but $= j^2 = -1$

$\therefore j^2w^2 = -(-1)w^2$

$= w^2$

$$\Rightarrow \frac{1}{2j}\left[\frac{2jw}{s^2+w^2}\right]$$

$$L\left[\sin wt\right] = \frac{w}{s^2+w^2} //$$

11) Find the laplace transform of $f(t) = \sin 2t$.

$$L(\sin 2t) = \int_{t=0}^{\infty} e^{-st} \cdot \sin 2t \, dt.$$

but $\sin 2t = \frac{1}{2j}\left[e^{jwt} - e^{-jwt}\right] = 1/2j \left(e^{2jt} - e^{-2jt}\right)$

$$\Rightarrow \int_{t=0}^{\infty} \frac{1}{2j}\left[e^{j2t} - e^{-j2t}\right] e^{-st} \, dt.$$

$$\Rightarrow \frac{1}{2j}\int_{t=0}^{\infty}\left(e^{2jt} \cdot e^{-st} - e^{-2jt} \cdot e^{-st}\right) dt.$$

$$\Rightarrow \frac{1}{2j}\int_{t=0}^{\infty} e^{-(s-2j)t} - e^{-(s+2j)t} \, dt.$$

$$\Rightarrow \frac{1}{2j}\left[\frac{e^{-(s-2)t}}{-(s-2)} - \frac{e^{-(s+2j)t}}{-(s+2j)}\right]_{t=0}^{\infty}$$

$$\Rightarrow \frac{1}{2j}\left(\frac{e^{-(s-2j)\infty}}{-(s-2j)} - \frac{e^{-(s+2j)\infty}}{-(s+2j)}\right]\left[\frac{e^{-(s-2j)0}}{(s-2j)} - \frac{e^{-(s+2j)0}}{(s+2j)}\right]$$

$$\Rightarrow \frac{1}{2j}\left[0+0+\frac{1}{(s-2j)} - \frac{1}{(s+2j)}\right]$$

$$\Rightarrow \frac{1}{2j}\left[\frac{s+2j-(s-2j)}{(s-2j)(s+2j)}\right]$$

$$\Rightarrow \frac{1}{2j}\left(\frac{s+2j-s+2j}{s^2+2js-2js-4j^2}\right]$$

$j^2 = -1, \; = -4(-1) = 4$ //

$$\Rightarrow \frac{1}{2j}\left[\frac{4j}{s^2+4}\right] = \frac{2}{s^2+4}$$

$$\mathcal{L}[\sin 2t] = \frac{2}{s^2+4} \;//$$

(2) Find the laplace tranform of $f(t) = \cos wt$.

$$\cos wt = \frac{1}{2}\left(e^{jwt} + e^{-jwt}\right)$$

$$\mathcal{L}(\cos wt) = \int_{t=0}^{\infty} e^{-st} \cdot \tfrac{1}{2}\left(e^{jwt} + e^{-jwt}\right) dt.$$

$$= \frac{1}{2}\int_{t=0}^{\infty} e^{-(s-jw)t} + e^{-(s+jw)t}\, dt.$$

$$\Rightarrow \frac{1}{2}\left[\frac{e^{-(s-jw)\infty}}{-(s-jw)} + \frac{e^{-(s+jw)\infty}}{-(s+jw)}\right] - \left[\frac{e^{-(s-jw)0}}{-(s-jw)} + \frac{e^{-(s+jw)0}}{-(s+jw)}\right]$$

$$\Rightarrow \frac{1}{2}\left[0+0+\frac{1}{(s-jw)}+\frac{1}{(s+jw)}\right] = \tfrac{1}{2}\left[\frac{s+jw+(s-jw)}{(s-jw)(s+jw)}\right]$$

$$\Rightarrow \frac{1}{2}\left[\frac{2s}{s^2+jws-jws-j^2w^2}\right]$$

but $j^2 = -1$

$\therefore\ -j^2w^2 = -(-1)w^2$

$$\Rightarrow \frac{1}{\cancel{2}}\left[\frac{2\cancel{s}}{s^2+w^2}\right] = \underline{\underline{\frac{s}{s^2+w^2}}}$$

$$\mathcal{L}\left[\cos wt\right] = \underline{\underline{\frac{s}{s^2+w^2}}}.$$

PARABOLIC FUNCTION

The function being given by $f(t) = t$, the laplace transform of this can be achieved as follows -

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$

Find the laplace transform of the followings:

1 t 2 t^2 3 t^3 4 t^4.

Solution

$$\mathcal{L}[t] = \frac{1!}{s^{1+1}} = \frac{1}{s^2}$$

$$\mathcal{L}[t^2] = \frac{2!}{s^{2+1}} = \frac{1}{s^3}$$

$$\mathcal{L}[t^3] = \frac{3!}{s^{3+1}} = \frac{3\times 2}{s^4} = \frac{6}{s^4}$$

$$\mathcal{L}[t^4] = \frac{4!}{s^{4+1}} = \frac{4\times 3\times 2}{s^5} = \frac{24}{s^5}$$

TABLE OF STANDARD TRANSFORMS

$F(t)$	$\mathcal{L} f[t] = F(s)$
a	a/s
e^{at}	$\frac{1}{s-a}$
$\sin at$	$\frac{a}{s^2+a^2}$
$\cos at$	$\frac{s}{s^2+a^2}$
$\sinh at$	$\frac{a}{s^2-a^2}$
$\cosh at$	$\frac{s}{s^2-a^2}$
t^n	$\frac{n!}{s^{n+1}}$

① Find the laplace transform of $f(t) = 2\sin 3t + \cos 3t$.

(15)

$\mathcal{L}[2\sin 3t + \cos 3t] = 2\mathcal{L}[\sin 3t] + \mathcal{L}[(\cos 3t)]$

$\Rightarrow 2 \cdot \frac{3}{s^2+9} + \frac{s}{s^2+9}$

$\Rightarrow \frac{2\times 3}{s^2+9} + \frac{s}{s^2+9} = \frac{6+s}{s^2+9}$ //

$\mathcal{L}[2\sin 3t + \cos 3t] = \frac{6+s}{s^2+9}$ //

② Find the laplace transform of $f(t) = 4e^{2t} + 3\cosh 4t$.

$\Rightarrow 4\mathcal{L}[e^{2t} + 3\mathcal{L}[\cosh 4]]$

$\Rightarrow 4 \cdot \frac{1}{s-2} + \frac{3 \cdot s}{s^2-4^2}$

$\Rightarrow \frac{4(s^2-16) + 3s(s-2)}{(s-2)(s^2-16)} \Rightarrow \frac{4s^2-64+3s^2-6s}{(s-2)(s^2-16)}$

$\Rightarrow \frac{7s^2-6s-64}{(s-2)(s^2-16)}$ // $\therefore \mathcal{L}[4e^{2t} + \cosh 4t] = \frac{7s^2-6s-64}{(s-2)(s^2-16)}$

(3) Find the laplace transform of $f(t) = 2\sin 3t + 4\sinh 3t$.

$$\Rightarrow 2[\sin] + 4\,\mathcal{L}(\sinh 3t)$$

$$\Rightarrow \frac{2\times 3}{s^2+9} + \frac{4\times 3}{s^2-9}.$$

$$\Rightarrow \frac{6}{s^2+9} + \frac{12}{s^2-9} \Rightarrow \frac{6(s^2-9)+12(s^2+9)}{(s^2+9)(s^2-9)}$$

$$\Rightarrow \frac{6s^2-54+12s^2+108}{(s^2-9)(s^2+9)} \Rightarrow \frac{6s^2+12s^2+108-54}{s^4-9s^2-9s^2-81}$$

$$= \frac{18s^2+54}{s^4-81}.$$

$$\mathcal{L}(2\sin 3t + 4\sinh 3t) = \frac{18s^2+54}{s^4-81}$$

(4) Find the laplace transform of $f(t) = 5e^{4t} + \cosh 2t$.

$$\Rightarrow 5\,\mathcal{L}[e^{4t}] + \mathcal{L}(\cosh 2t)$$

$$\Rightarrow 5\cdot\frac{1}{(s-4)} + \frac{s}{(s^2-4)} \Rightarrow \frac{5(s^2-4)+s(s-4)}{(s-4)(s^2-4)}$$